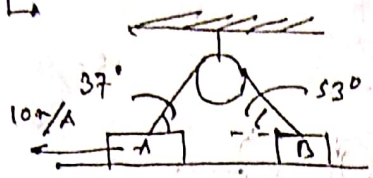
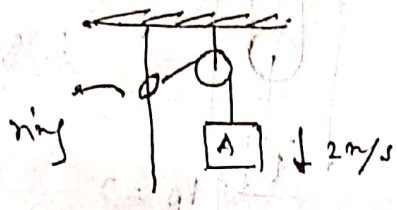
Constrained Motion

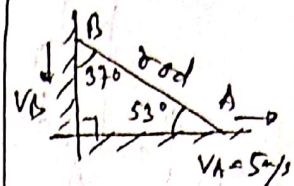
① String Constraint & Pulley Constraint :-



$V_B = ?$ (13.33 m/s)

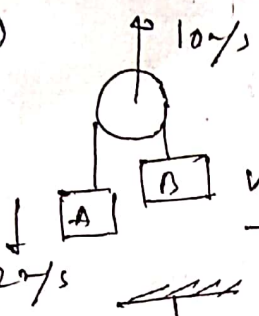
② 

V_{ring} ? (2.5 m/s)

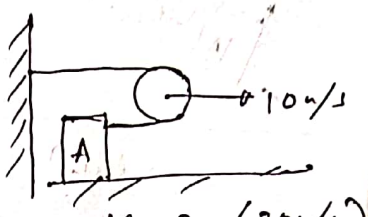
③ 

(velo of end A)

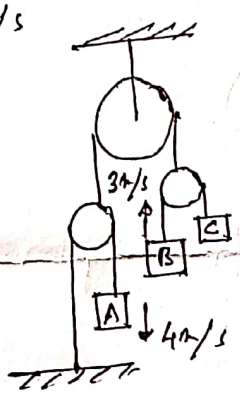
find V_A (velocity of end B) ? (3.75 m/s)

④ 

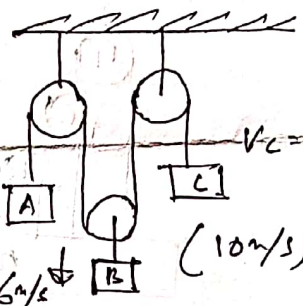
$V_B = ?$ (22 m/s up)

⑤ 

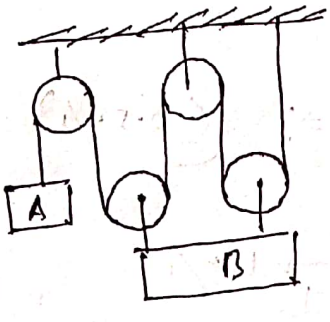
V_A ? (20 m/s)

⑤ 

$V_C = ?$ (1 m/s, up)

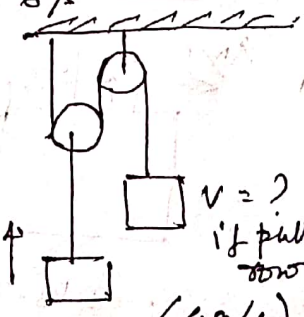
⑥ 

(10 m/s, up)

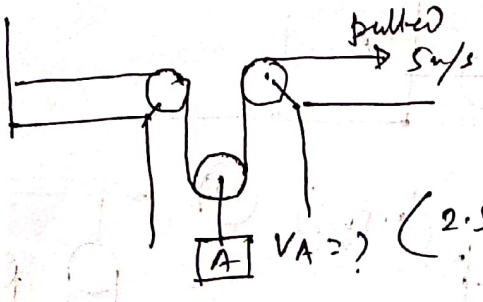
⑦ 

(1 m/s up)

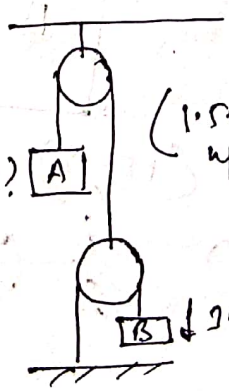
$V_B = ?$

⑧ 

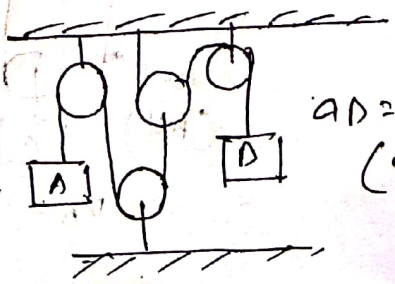
(4 m/s)

⑨ 

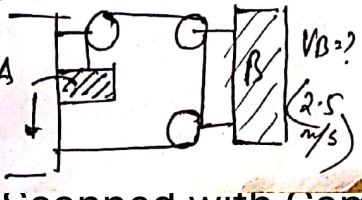
$V_A = ?$ (2.5 m/s up)

⑩ 

$V_A = ?$ (1.5 m/s) up

⑪ 

$a_B = ?$ ($2 m/s^2$ up)

⑫ 

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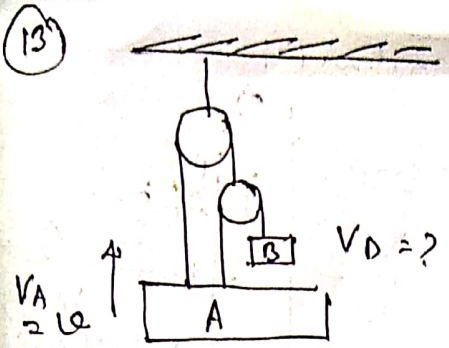
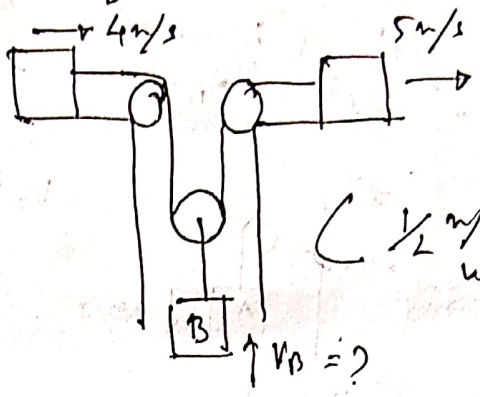
(13)



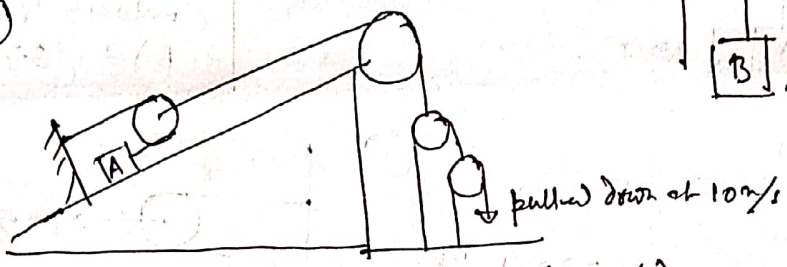
$V_D = ?$ (3v, down)

(14)



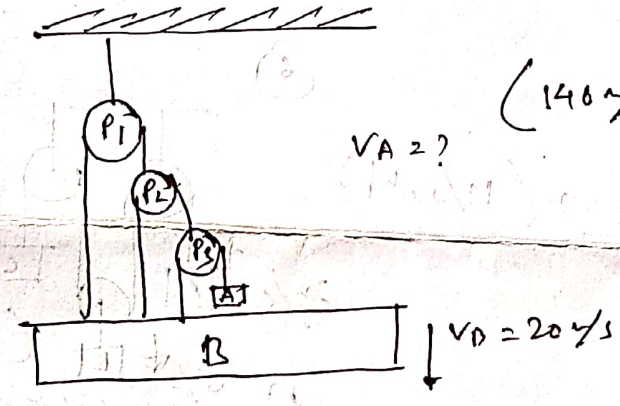
(1/2 m/s up)

(15)



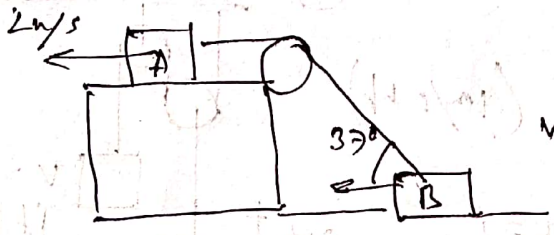
find $V_A = ?$ (5m/s)

(16)



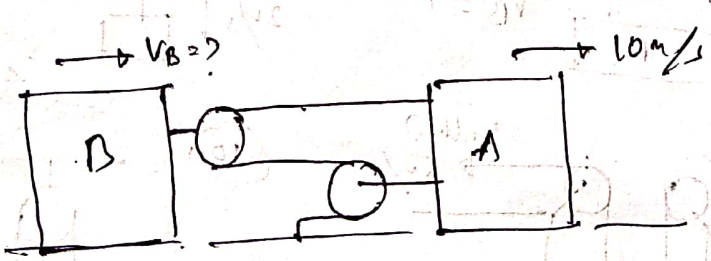
$V_A = ?$ (140 m/s)

(17)



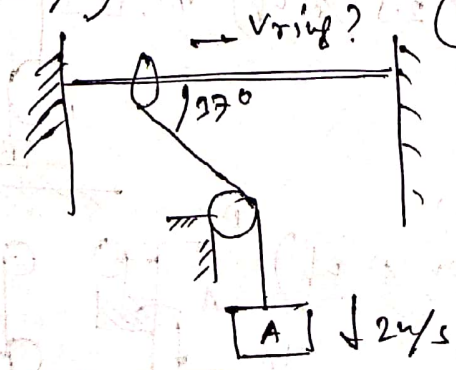
$V_D = ?$ (2.5 m/s)

(18)



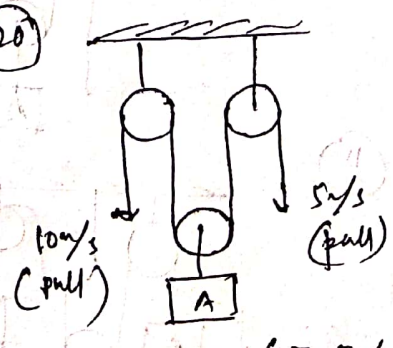
(15m/s)

(19)



(2.5m/s)

(20)



$V_A = ?$ (7.5m/s up)

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